Meter Chalu: build spec

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Meter Chalu is a hands-free voice coach that helps auto and taxi drivers in Maharashtra practise everyday Marathi, with help in their own language. We demo by Nov 8 and submit to the Gnani AI challenge before 11:59 PM IST on Nov 10.

What we're building

Name: Meter Chalu. **Tagline:** "भाषा चालू, मीटर चालू!" (Get your language going, and your meter keeps running.)

Problem. Since August 2026, Maharashtra requires auto, taxi and app-cab drivers to have working Marathi. Non-compliance means licence suspension of up to 3 months, then cancellation. RTOs have screened 8,217 drivers and issued 1,268 notices (Open, Outlook). The check is spoken, and drivers on 12-hour shifts can't attend classes.

Users. Auto, taxi and app-cab drivers in Maharashtra, many of them migrants who speak Hindi or another Indian language, often with low literacy.

Core idea. The app plays a Marathi-speaking passenger. The driver answers by voice in whatever he can manage. A helper called **Saathi** explains, corrects and answers doubts in the driver's own language, then the ride continues.

Decisions locked

Topic	Decision
Language taught	Marathi only. No other learning languages.
Helper language	Driver picks it. Hindi is the default and must be perfect; Bengali, Telugu, Tamil, Kannada, Gujarati, Punjabi, Malayalam and English are options. Demo shows Hindi fully plus one short clip in another language.
Screen text	Shown in the chosen helper language.
Login	None. Name and helper language are saved on the phone on first launch.

Topic	Decision
Mic	One tap starts the session; the mic stays on (hands-free).
Readiness	One overall readiness meter, labelled as a practice score, not a certificate.
Scaling story	Within Maharashtra (Mumbai, then Pune, Nagpur, Nashik), not other languages.
Framing	Neutral on the rule: "whatever you think of it, drivers' livelihoods depend on it." Never claim to be official.
Data	Synthetic data only. No real phone numbers, Aadhaar or recorded calls. Consent from anyone filmed.

App flow

First launch only

1. Splash (about 2 s): logo, "Meter Chalu", tagline, auto and taxi visuals. Saathi says "भाषा चालू, मीटर चालू!"
2. Saathi asks the helper language (tap a card or speak). Hindi is pre-selected.
3. Saathi asks the driver's name. Both are saved on the phone and never asked again (editable in Settings).
4. Home screen.

Every launch after that

1. Splash (2 s or less).
2. Home: "नमस्ते रमेश भाई!", the readiness meter, and one big "Start practice" button.
3. Practice session (next section).
4. Summary screen, then back to Home.

Practice session

The driver taps Start once; everything after that is by voice until he stops.

1. **Choose the situation.** Saathi (helper language) offers: station ride, directions, fare and change, elderly passenger, airport and luggage, police or RTO check (mock), or "tell me your own situation". For a custom one, Gemini turns his description into a passenger role and 4 to 6 steps.
2. **Passenger speaks** one line in Marathi. The screen shows it big, with the meaning small underneath.
3. **Driver replies** in mixed Marathi and his own language.
4. **Saathi's short analysis** (helper language, about 10 to 15 s): what was right, what to change, the correct Marathi sentence spoken slowly, then "अब आप बोलिए".
5. **Driver repeats** the correct line; Saathi confirms. He can ask any doubt here; "aur samjhao" gets the detailed version (word meanings, a grammar hint, a politeness tip, another way to say it).
6. **Resume:** "चलिए, आगे बढ़ते हैं" and the next passenger line. Steps 2 to 6 repeat until the ride ends.
7. **End:** the summary screen.

Every Saathi turn ends by telling the driver what to do next, so it's always clear whose turn it is. Unclear audio gets "आवाज़ साफ़ नहीं आई, फिर बोलिए" instead of a wrong analysis.

Voice commands

The driver can say these anytime, in any helper language. Each also has an on-screen button where noted.

He says	Saathi does	Button
"फिर से" / "repeat"	Repeats the last line	Replay
"धीरे बोलो"	Repeats it slowly	
"मतलब?" / "समझ नहीं आया"	Explains the line in his language	मतलब?
"Marathi mein ___ kaise bolte hain?"	Teaches the phrase, he repeats, Saathi checks	
"aur samjhao" / "kyun?"	Detailed explanation	
"दूसरी situation"	Back to choosing a situation	
"रुको"	Pauses (a real passenger arrived)	Pause

He says	Saathi does	Button
"बस" / "बंद करो" / "thamba"	Ends the session, shows the summary	Stop
(silent about 8 s)	Offers a hint	

Feature list

Must-have for the demo

- [ ] Splash, first-launch name and helper language, saved on the phone
- [ ] One-tap hands-free session (LiveKit agent)
- [ ] 6 general situations plus "your own situation"
- [ ] Short analysis, correct sentence, repeat-after-me, doubts, resume
- [ ] Voice commands: repeat, slow, meaning, change situation, pause, stop
- [ ] Ride progress bar inside the session
- [ ] One readiness meter on Home
- [ ] Summary: "आज के वाक्य" with replay, plus one mistake to work on
- [ ] Listening in Marathi plus the helper language at once (see gotchas)

Nice-to-have if time allows

- [ ] Next session starts with 2 phrases he got wrong last time
- [ ] Quick phrasebook: 20 most-used ride phrases with audio, works offline
- [ ] Numbers and fares drill (1 to 500)
- [ ] Easy mode and RTO-mock mode (less help)
- [ ] Daily streak
- [ ] WhatsApp share

Skip: login or OTP, leaderboards, typing exercises, written grammar lessons, other learning languages.

New ideas for outreach and impact

Ranked by how much they help the driver and how far they spread. The first three are cheap enough to do before Nov 10.

1. **"मी मराठी शिकतोय" auto sticker.** A printable yellow sticker for the back of the auto: "I'm learning Marathi, speak with me!" plus a QR to the app. Passengers start talking Marathi with the driver, which is free practice, and every auto becomes an ad. Great visual for the video.
2. **Hashtag campaign #MeterChaluMarathiChalu.** Ask viewers to say "शाबास!" to a driver trying Marathi. It reframes a divisive rule into encouragement, and positive posts get shared more.
3. **Shareable progress card.** At 50% and 100% readiness, generate an image ("रमेश भाई: 50% तैयार") to send on WhatsApp. Driver WhatsApp groups are the real distribution channel. Label it as a practice score, not a certificate.
4. **WhatsApp voice-note mode.** The driver sends a voice note and gets a voice reply. No app install, and drivers already live on WhatsApp. Good "next step" for the pitch.
5. **Phone-call mode for basic phones.** Call a number and practise, using Gnani's 8 kHz telephony strength and LiveKit phone support. Strong for the noisy-audio award and the interview.
6. **Union and stand partnerships.** Pune unions already run Marathi classes at auto stands. Pitch the app as homework between classes; a stand-wide QR poster reaches dozens of drivers at once.
7. **Community situation library.** Custom situations drivers describe (anonymised) become new practice rides for everyone, e.g. "AC taxi chahiye".
8. **Impact counter in the app and post.** Show honest totals: sessions practised, phrases learned. Real numbers make a stronger post than claims.

UI guidelines

Screens: Splash → (first launch: language, name) → Home → Situation → Ride practice → Summary.

Ride practice screen (the one in the video): auto icon moving toward a flag (ride progress), the Marathi line big with its meaning small below, the driver's transcribed reply, a green tip card, a large pulsing "listening" indicator, and buttons for Replay, मतलब?, Pause and Stop.

Rules

- Icons and audio before text: many users have low literacy.
- Touch targets 56 px or larger, anchored at the bottom for one-handed use.
- Auto-rickshaw yellow and black, high contrast, readable in sunlight.

- Text 20 px or larger; the Marathi line is the biggest thing on screen.
- No red crosses. Use yellow for "almost" and green for "great".
- Always show the state: listening, thinking, speaking.
- Every screen in the chosen helper language.
- Works on a cheap Android phone on mobile data.
- Splash 2 s or less; avoid readable number plates or faces in photos unless you have consent.

Build route: restyle LiveKit's React starter (agent-starter-react) to these screens; it connects to `agent.py` via `python agent.py dev`.

Tech stack and gotchas

Layer	Choice
Speech-to-text	Gnani Prisma, live streaming via LiveKit plugin `livekit-plugins-gnani`
Text-to-speech	Gnani Timbre v2.5: Marathi voice Zahira; helper voices Nalini (Hindi), Kirra (Bengali), Suhana (Telugu), Asmita (Tamil), Saanvi (Kannada), Falak (Gujarati), Mehuli (Punjabi), Reshma (Malayalam), Kaveri (English)
LLM	Gemini 3.5 Flash, auto-fallback to 3-flash-preview and 3.8 Flash
Voice pipeline	LiveKit Agents 1.8 (`agent.py`), Silero voice detection
Frontend	LiveKit React starter, restyled

Already built (`agent.py`): a menu agent (language and situation by voice), a ride agent that speaks tips in the helper voice and passenger lines in the Marathi voice, 6 situations, a back-to-menu tool, and the Gemini fallback chain. Run with `python agent.py console`. Not yet tested end to end by voice.

Gotchas found so far

- Live STT accepts one language unless the model is Prisma v3.0. The plugin doesn't send the `model` header, so it's locked to `mr-IN`. Fix: subclass the plugin's stream to add `model: prisma-v3.0` and `language_code: mr-IN,<helper code>` (Gnani allows up to 4).
- The plugin only sends `language` to Timbre v2.5, so use v2.5 voices.
- Gemini needs API version `v1beta` (system instructions) and a 30 s timeout. Gemini 2.5 Flash is closed to new users. 3.8 Flash sometimes returns 503/504; the fallback handles it.
- HTTP 401 from Gnani = bad key, wrong env file, or no credits. Set `ENV_FILE` at the top of `agent.py`.
- Credits: two voices per reply uses more Gnani credits. Cache fixed lines (openings, menu prompts) where possible.
- Free Gemini tier is limited (about 20 requests a day reported for 3.8 Flash).

Demo, posting and submission

Video (about 2:30, plus a 60 s cut for the Demo Day Drop award)

Time	Section
0:00–0:08	Hook: "No Marathi = no licence" headlines
0:08–0:35	Problem: drivers, 12-hour shifts, spoken test, RTO notices
0:35–0:50	Solution: Meter Chalu, "भाषा चालू, मीटर चालू!"
0:50–1:50	Real demo with a consenting driver in an auto, phone screen picture-in-picture
1:50–2:10	How it works: Gnani Prisma → Gemini → Gnani Timbre, on LiveKit
2:10–2:25	Impact and next: Pune, Nagpur, Nashik; phone-call mode
2:25–2:35	Call to action: share with one driver you know

Captions on everything, cut every pause, yellow-and-black title cards, one honest limitation shown. Tools: CapCut or DaVinci Resolve, Canva for cards.

Timeline

- [ ] Nov 7: core loop working by voice, UI started
- [ ] Nov 8: UI done, film, edit, post on LinkedIn and X with #GnaniAI #GreatIndianAIInternshipChallenge and tag Gnani AI
- [ ] Nov 9: share in groups, reply to every comment, fix bugs
- [ ] Nov 10: submit before 11:59 PM IST; nominate the posts in step 4

Submission form fields: project title, problem, who will use it, Gnani models used (Prisma, Timbre), each member's contribution.

Open questions: does Gnani keep counting engagement after Nov 10? Does the form require a public repo? Ask on Discord.